

PROJECT PRESENTATION

CareerPath AI

Intelligent Resume Analysis & Career Planning Platform

Democratizing career guidance for students and fresh graduates through AI

Challenges in Early Careers

PROBLEM STATEMENT



ATS Incompatibility

Poor resume quality and missing structure cause up to 75% of potential applications to be auto-rejected by ATS systems.



Unclear Pathways

Students and fresh graduates face severe delays in early career choices due to unclear individual fit with roles.



Skill Awareness Gaps

Lack of structured insight on technical & behavioral requirements makes prep for target roles highly ineffective.

Platform Value Proposition

EXECUTIVE OVERVIEW

🎯 Targeting Fresh Talents

Specially crafted for students, early-career seekers, and fresh graduates aiming to optimize their digital profile and direct their job hunt effectively with quantitative indicators.

📈 Strategic Growth Impact

Accelerates placement rates, reduces job searching cycles, and provides data-backed learning paths that efficiently bridge professional capabilities and market demand.

Technical Infrastructure

TECHNOLOGY STACK



Frontend Interface

Implemented in robust Python-based Streamlit, allowing fast real-time user-model interactions and reactive visual elements.



NLP Parsing Core

Powered by spaCy Named Entity Recognition pipelines trained particularly on modern tech and job-market vocabulary.



ML & Predictors

Utilizes Scikit-Learn algorithms alongside Pandas, NumPy, and Plotly to process and map structured professional features.

System Processing Pipeline

SYSTEM PIPELINE

- ➔ **User Profile Ingestion:** Accepts and converts raw multi-format files (PDF/DOCX) into normalized programmatic content.
- ➔ **Custom NER Parsing:** Employs custom spaCy pipeline models to extract detailed user skills, work fields, and background assets.
- ➔ **Inference Engines:** Triggers concurrent specialized engines (ATS Scorer, Role Classifier, and Gap Analyzer).
- ➔ **Dynamic Dashboard Feed:** Streams structural matching, personalized roadmaps, and confidence factors to the user interface.

Resume Compatibility Engine

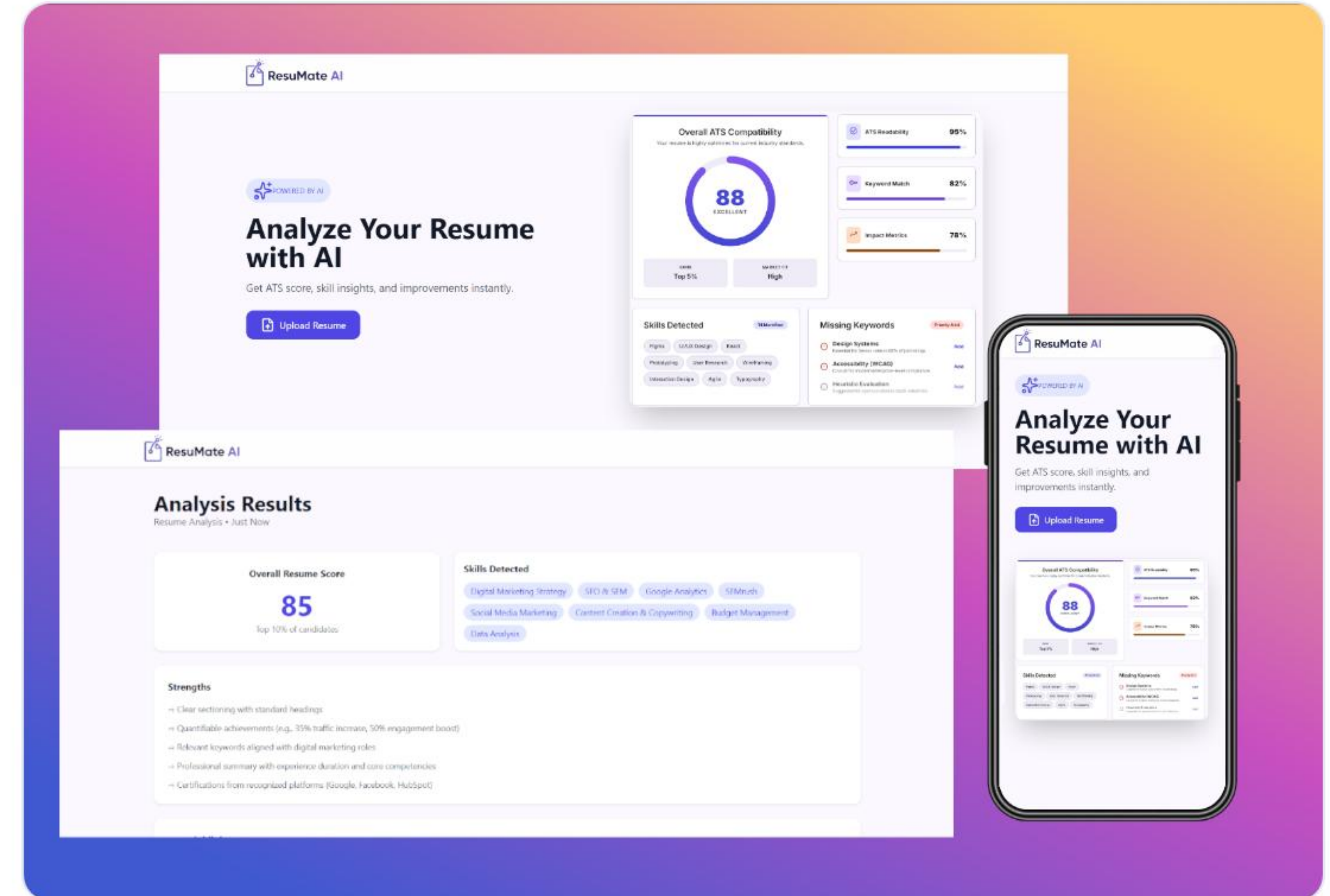
CORE FEATURE

Intelligent Optimization Feedback

Evaluates candidate documents instantly against standard Applicant Tracking Systems (ATS) formatting constraints.

Provides a granular 0–100 quality scoring gauge combined with targeted tips to enrich keywords and structural balance.

Delivers recommendations focused on maximizing resume scannability across enterprise filters.



Salary Prediction Factors

ESTIMATION LOGIC

Input Factor	Model Weight	Strategic Relevance
Experience Level	30%	Calculates professional depth and years spent in relevant industries.
Technical Skills	25%	Weights in-demand technical languages, cloud setups, and specialized tools.
Location Coordinates	20%	Adjusts output according to geographic compensation shifts.
Company Size	15%	Normalizes rates based on business scale (enterprise scale vs early startups).
Education Profile	10%	Weights formal degrees, technical certifications, and academic levels.

Advanced Interview Prep

PREPARATION TOOLSET



AI Mock Questioning

Generates tailored conversational and core role questions directly linked to target sectors.



Answer Evaluation

Systematically parses answer attempts, highlighting structural weaknesses and giving recommendations.



Behavioral Coaching

Uses built-in coaches to tutor applicants on the STAR methodology for high-impact communication.

Codebase Architecture

IMPLEMENTATION

Main Application Tree

CareerPath_AI/

- |— main.py (Streamlit Entry)
- |— requirements.txt (Dependencies)
- └─ models/ (Serialized PKL Predictors)
 - |— ats_scorer.pkl
 - |— salary_predictor.pkl

Parsing Utilities

utils/

- |— resume_parser.py (spaCy Custom NER)
- |— feature_extractor.py (Embeddings)
- └─ recommendations.py (Logic Engine)

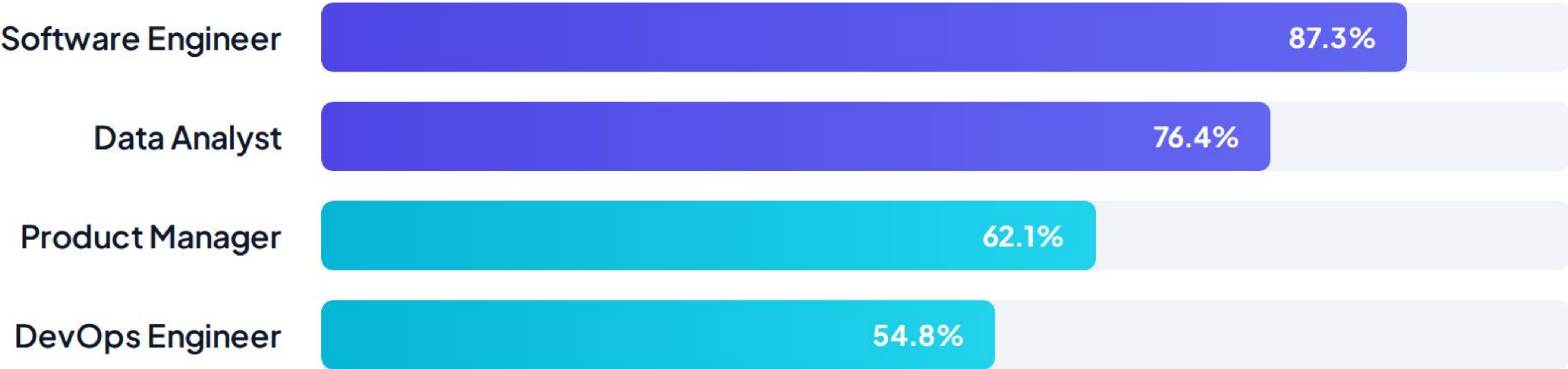
Data & Model Lifecycle

DEEP DIVE

Development Stage	Specific Purpose	Integrated Tech Stack
Collection	Gathers training portfolios from real resumes & postings.	Web Scraping pipelines
Preprocessing	Applies text normalization and tokenization logic.	spaCy & NLTK
Feature Engineering	Converts natural text properties into numerical vectors.	TF-IDF / Word2Vec models
Inference Testing	Evaluates classification and prediction models on test splits.	Scikit-Learn estimators
System Deployment	Streams reactive predictions directly to web views.	Streamlit & Pickle formats

Role Confidence Metrics

LIVE ANALYTICS



The figure illustrates classification confidence distributions for a sample profile, highlighting targeted alignment and fit probability.

Platform Metrics

EXPERIMENTAL VALIDATION

Analytical Engine Module	Model Accuracy	Precision Rate	Recall Score	F1 Evaluation
Job Role Classifier	87.3%	85.6%	86.8%	86.2%
ATS Score Validator	91.2%	89.5%	92.1%	90.8%
Salary Predictor (RMSE)	\$8,450	N/A	N/A	N/A
Company Recommender (MAP@10)	78.4%	N/A	N/A	N/A

Team Responsibilities

COORDINATION

Technical Backends

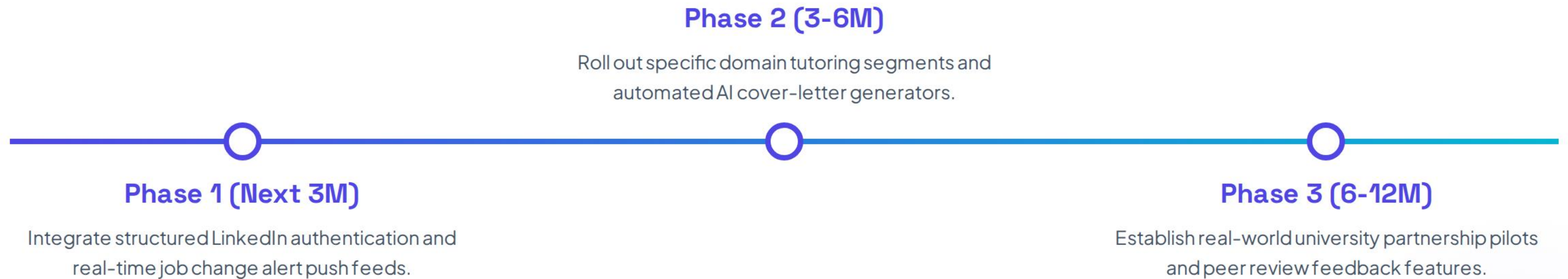
Coordinated by our ML Engineers and NLP Specialists to drive robust custom entity recognition pipelines, parse raw formats cleanly, and train highly accurate predictive salary models.

Dashboard & Analytics

Guided by our UI/UX Designers, Web Developers, and Analysts to manage interface responsive behaviors, render performance statistics, and display dynamic timeline graphs.

System Growth Roadmap

SYSTEM ROADMAP



THANK YOU

Image Sources



Thumbnail
for
github.com

<https://repository-images.githubusercontent.com/1222967037/0446611f-f859-42d3-b1a5-5c65f732e5d9>

Source: github.com